

Task 1

1) Finding the error in the data and fix it (missing data “lipids 5”):

R code:

```
sum(is.na(data$lipids5))  
mean(is.na(data$lipids5)) * 100
```

Output:

```
> sum(is.na(data$lipids5))  
[1] 276  
  
> mean(is.na(data$lipids5)) * 100  
[1] 24.04181
```

The variable lipids5 was examined for data quality issues. Missing values were identified (276), indicating incomplete observations in the dataset.

R code:

```
data$lipids5[is.na(data$lipids5)] <- median(data$lipids5, na.rm = TRUE)  
sum(is.na(data$lipids5))
```

Output:

```
> data$lipids5[is.na(data$lipids5)] <- median(data$lipids5, na.rm = TRUE)  
> sum(is.na(data$lipids5))  
[1] 0
```

The variable lipids5 contained missing values, which were identified using standard data inspection methods. To address this issue, median imputation was applied, replacing missing values with the median of the observed data. This approach was chosen because the variable exhibits skewness, and the median provides a robust measure that is less sensitive to extreme values.

2) Load and inspect the dataset:

The dataset was loaded into R and its structure was examined to identify variable types and ensure correct data handling. Special attention was given to distinguishing continuous variables from categorical ones.

R code:

```
library(MASS)

library(dplyr)

data <- read.csv("/Users/apple/Downloads/INRTU courses/2nd semester/Data
Analysis/data_for_analysis.csv", header = T)

str(data)
```

Output:

```
'data.frame':    1148 obs. of  31 variables:
 $ record_id      : int  1 2 3 4 6 8 9 10 12 13 ...
 $ outcome        : int  0 0 0 0 0 1 0 1 1 0 ...
 $ factor_eth     : int  1 1 1 1 1 1 1 1 1 2 ...
 $ factor_h       : int  0 0 0 0 0 0 0 1 0 1 ...
 $ factor_pcos    : int  NA NA NA NA NA NA NA NA NA 1 ...
 $ factor_prl     : int  0 0 0 0 0 0 0 0 0 0 ...
 $ hormone1       : num  0.5 1.9 1.2 3.8 1.6 1.6 0.42 2.7 1 1.8 ...
 $ hormone2       : int  314 196 311 409 418 309 415 450 198 569 ...
 $ hormone3       : num  9.9 6.8 1.2 3.2 3.8 2.7 5.9 7 9 20.6 ...
 $ hormone4       : num  4.2 6.8 3.2 2.7 7.2 1.6 9.1 4.5 5.2 8.3 ...
 $ hormone5       : num  322.7 190.9 153.7 225.7 99.9 ...
 $ hormone6       : num  166.1 41 140.8 93.4 24.7 ...
 $ hormone7       : num  5.7 6.8 5.4 12.7 1.6 9.9 7.5 4 4.2 1.7 ...
 $ hormone8       : num  129 260 81.1 148 121 120 135 199 149 238 ...
 $ hormone10_generated: num  0.922 0.166 0.405 0.441 0.199 ...
 $ lipids1        : num  0.4 0.54 0.45 0.38 0.48 0.34 0.49 0.62 1.1 0.51
 ...
 $ lipids2        : num  5.05 3.9 3.5 4.84 3.96 3.85 3.93 4.3 3.89 3.72
 ...
 $ lipids3        : num  1.23 1.67 1.29 1.32 1.85 1.42 1.26 1.3 1.11 0.78
 ...
 $ lipids4        : num  3.64 1.98 2.01 3.35 1.89 2.28 2.45 2.72 2.28
 2.28 ...
```

```

$ lipids5          : num  0.18 0.25 0.2 0.17 0.22 0.15 0.22 0.28 0.5 NA
...
$ carb_metabolism  : num  5.41 4.5 5.57 5.95 4.15 4.66 4.44 5.32 4.71 4.39
...
$ lipid_pero1      : num  2.36 2.82 2.16 2.44 1.98 2.36 3.06 2.48 1.83
3.62 ...
$ lipid_pero2      : num  1.56 1.36 0.72 1.86 0.86 1.17 1.64 1.22 1.35
1.26 ...
$ lipid_pero3      : num  1.14 0.71 0.27 1.46 0.33 0.62 0.76 0.81 0.65
0.43 ...
$ lipid_pero4      : num  0.87 1.51 0.64 1.32 2.04 2.61 1.77 0.58 0.39
2.02 ...
$ lipid_pero5      : num  10.4 13.7 10.6 10.4 15.1 ...
$ antioxidant1     : num  1.64 1.33 1.85 1.34 1.72 1.43 1.7 1.31 1.58 1.73
...
$ antioxidant2     : num  2.24 2.48 2.19 1.58 2.14 2.14 2.22 2.02 2.15
2.36 ...
$ antioxidant3     : num  1.91 1.98 1.92 1.86 2.34 1.83 2.08 1.98 2.24
2.11 ...
$ antioxidant4     : num  5.21 5.06 9.58 7.46 9.22 7.62 13 7.8 8.85 8.4
...
$ antioxidant5     : num  0.38 0.39 0.49 0.43 0.86 0.38 0.94 0.47 1.12
0.74 ...

```

R code:

```
Summary(data)
```

Output:

record_id	outcome	factor_eth	factor_h	factor_pcos
factor_prl				
Min. : 1	Min. :0.0000	Min. :1.00	Min. :0.000	Min. :1.0
Min. :0.00000				
1st Qu.: 655	1st Qu.:0.0000	1st Qu.:1.00	1st Qu.:0.000	1st Qu.:1.0
1st Qu.:0.00000				
Median :1258	Median :0.0000	Median :1.00	Median :0.000	Median :3.0
Median :0.00000				
Mean :1306	Mean :0.1395	Mean :1.46	Mean :0.194	Mean :2.6
Mean :0.06708				
3rd Qu.:1946	3rd Qu.:0.0000	3rd Qu.:2.00	3rd Qu.:0.000	3rd Qu.:4.0
3rd Qu.:0.00000				

Max. :2694	Max. :1.0000	Max. :3.00	Max. :1.000	Max. :4.0
Max. :1.00000				
NA's :15	NA's :1	NA's :1	NA's :14	NA's :983
hormone1	hormone2	hormone3	hormone4	
hormone5				
Min. : 0.050	Min. : 31.0	Min. : 0.000	Min. : 0.100	Min. : 0.0
1st Qu.: 1.100	1st Qu.: 227.0	1st Qu.: 3.500	1st Qu.: 3.800	1st Qu.: 179.3
Median : 1.500	Median : 317.0	Median : 5.700	Median : 5.500	Median : 270.5
Mean : 1.801	Mean : 380.2	Mean : 9.115	Mean : 7.771	Mean : 327.5
3rd Qu.: 2.100	3rd Qu.: 457.0	3rd Qu.: 9.500	3rd Qu.: 7.400	3rd Qu.: 378.8
Max. :26.100	Max. :2798.0	Max. :118.400	Max. :131.400	Max. :5686.8
hormone6	hormone7	hormone8	hormone10_generated	
lipids1				
Min. : 0.76	Min. : 0.000	Min. : 16.3	Min. : 0.0770	Min. :0.2000
1st Qu.: 40.58	1st Qu.: 2.600	1st Qu.:115.8	1st Qu.: 0.2860	1st Qu.:0.6875
Median : 64.85	Median : 5.100	Median :162.0	Median : 0.4310	Median :0.9200
Mean : 78.45	Mean : 5.628	Mean :176.3	Mean : 0.7258	Mean :0.9610
3rd Qu.:100.97	3rd Qu.: 7.600	3rd Qu.:218.0	3rd Qu.: 0.6380	3rd Qu.:1.1800
Max. :375.20	Max. :47.100	Max. :558.0	Max. :261.0000	Max. :2.7300
lipids2	lipids3	lipids4	lipids5	
carb_metabolism				
Min. :2.430	Min. :0.410	Min. :0.820	Min. :0.0900	Min. :3.040

1st Qu.:3.840 4.300	1st Qu.:0.920	1st Qu.:2.330	1st Qu.:0.3000	1st Qu.:
Median :4.390 4.800	Median :1.070	Median :2.840	Median :0.4200	Median :
Mean :4.504 4.889	Mean :1.106	Mean :2.966	Mean :0.4333	Mean :
3rd Qu.:5.040 5.380	3rd Qu.:1.270	3rd Qu.:3.442	3rd Qu.:0.5400	3rd Qu.:
Max. :8.820 :11.500	Max. :2.010	Max. :7.010	Max. :1.2400	Max.
			NA's :276	
lipid_pero1 lipid_pero5	lipid_pero2	lipid_pero3	lipid_pero4	
Min. :0.500 1.750	Min. :0.120	Min. :0.0700	Min. : 0.23	Min. :
1st Qu.:1.510 9.277	1st Qu.:0.980	1st Qu.:0.4075	1st Qu.: 0.93	1st Qu.:
Median :1.980 :11.725	Median :1.300	Median :0.6000	Median : 1.33	Median
Mean :2.084 :11.864	Mean :1.393	Mean :0.6916	Mean : 1.44	Mean
3rd Qu.:2.562 Qu.:14.360	3rd Qu.:1.700	3rd Qu.:0.9125	3rd Qu.: 1.80	3rd
Max. :5.020 :27.800	Max. :3.700	Max. :1.9300	Max. :12.40	Max.
antioxidant1 antioxidant5	antioxidant2	antioxidant3	antioxidant4	
Min. :1.070 :0.2300	Min. :1.18	Min. :0.940	Min. : 0.000	Min.
1st Qu.:1.480 Qu.:0.4800	1st Qu.:2.00	1st Qu.:1.730	1st Qu.: 5.460	1st
Median :1.630 :0.6200	Median :2.31	Median :1.950	Median : 7.340	Median
Mean :1.621 :0.6922	Mean :2.34	Mean :1.973	Mean : 7.853	Mean
3rd Qu.:1.760 Qu.:0.8400	3rd Qu.:2.61	3rd Qu.:2.180	3rd Qu.: 9.680	3rd

Max. :2.430	Max. :3.82	Max. :3.590	Max. :25.000	Max. :2.0400
-------------	------------	-------------	--------------	--------------

The dataset contains both discrete and continuous variables. The variable outcome is used to define groups, while numerical variables represent hormone levels, indices, and other measurements.

Select continuous variables:

The dataset contains both continuous and discrete variables. Although many variables are stored in numeric format, not all of them represent continuous measurements. Therefore, variables corresponding to categorical or discrete factors, such as factor_eth, factor_h, factor_pcos, and factor_prl, hormone2, were excluded from the analysis. Only true continuous variables were retained for distribution fitting.

R code:

```
continuous_vars <- data %>%
  select(where(is.numeric)) %>%
  select(-outcome, -record_id,
         -factor_eth, -factor_h, -factor_pcos, -factor_prl, -hormone2) %>%
  colnames()
continuous_vars
```

Output:

```
[1] "hormone1"      "hormone3"      "hormone4"
     "hormone5"
[5] "hormone6"      "hormone7"      "hormone8"
     "hormone10_generated"
[9] "lipids1"       "lipids2"       "lipids3"
     "lipids4"
[13] "lipids5"       "carb_metabolism" "lipid_pero1"
     "lipid_pero2"
[17] "lipid_pero3"   "lipid_pero4"   "lipid_pero5"
     "antioxidant1"
[21] "antioxidant2"  "antioxidant3"  "antioxidant4"
     "antioxidant5"
```

A set of continuous variables was obtained, which will be analyzed separately for each outcome group.

Split data by outcome:

The dataset was divided into groups based on the outcome variable to allow group-wise distribution analysis.

R code:

```
group0 <- data %>% filter(outcome == 0)
group1 <- data %>% filter(outcome == 1)
```

This separation allows comparison of distributional properties between individuals with and without the outcome.

2) Fit distributions:

To illustrate the procedure, the variable *hormone1* is analyzed. Three candidate distributions are fitted using maximum likelihood estimation.

R code:

```
x <- na.omit(group0$hormone1)
fit_norm <- fitdistr(x, "normal")
fit_lnorm <- fitdistr(x[x > 0], "lognormal")
fit_exp <- fitdistr(x[x > 0], "exponential")
BIC_value<- c(BIC(fit_norm), BIC(fit_lnorm), BIC(fit_exp))
Distributions<- c("Normal", "Log-Normal", "Exponential")
result<-data.frame(BIC_Value=BIC_value, Distributions=Distributions)
result<- result[order(result$BIC_Value, decreasing = F),]
result
```

Output:

	BIC_Value	Distributions
2	2634.004	Log-Normal
3	3167.093	Exponential
1	3716.618	Normal

From the output we can see that the Log-Normal distribution has the LOWEST BIC value, making it the best fit for *hormone1*.

The same procedure was applied to all continuous variables for both outcome groups. For each variable, normal, log-normal, and exponential distributions were fitted using maximum likelihood estimation. The Bayesian Information Criterion was computed using the BIC() function, and the distribution with the lowest BIC value was selected as the best-fitting model.

	variable	group	best_distribution	BIC_normal	BIC_lognormal	BIC_exponential
1	hormone1	0	lognormal	3716.61795	2634.00391	3167.09257
2	hormone1	1	lognormal	415.85733	367.17627	486.89516
3	hormone3	0	lognormal	7649.05142	6075.33054	6311.82052
4	hormone3	1	lognormal	1252.73822	1030.05619	1059.02955
5	hormone4	0	lognormal	7533.14912	5432.53924	5960.33886
6	hormone4	1	lognormal	1294.53807	965.40651	1042.14482
7	hormone5	0	lognormal	14274.62582	12966.35772	13263.51170
8	hormone5	1	lognormal	2210.87979	2035.95353	2101.59727
9	hormone6	0	lognormal	10687.61206	10253.87433	10603.42812
10	hormone6	1	lognormal	1706.52048	1660.27375	1709.87396
11	hormone7	0	lognormal	5499.75806	5169.75644	5383.59369
12	hormone7	1	lognormal	904.38440	865.11011	879.20495

	variable	group	best_distribution	BIC_normal	BIC_lognormal	BIC_exponential
13	hormone8	0	lognormal	11634.69519	11453.69605	12232.01481
14	hormone8	1	lognormal	1827.10195	1811.24932	1936.40612
15	hormone10_generated	0	lognormal	6990.85411	216.56711	1443.62305
16	hormone10_generated	1	lognormal	141.25920	38.96245	108.50657
17	lipids1	0	lognormal	940.87040	818.46020	1894.76259
18	lipids1	1	lognormal	145.42329	143.90368	320.11696
19	lipids2	0	lognormal	2629.10555	2539.44738	4932.94155
20	lipids2	1	lognormal	450.91013	432.81162	824.59383
21	lipids3	0	lognormal	277.82240	223.63444	2176.64048
22	lipids3	1	lognormal	39.99264	28.05805	359.90858
23	lipids4	0	lognormal	2549.08368	2458.57449	4101.57013
24	lipids4	1	lognormal	445.09381	427.38661	697.07712
25	carb_metabolism	0	lognormal	2389.43229	2272.48393	5116.05621
26	carb_metabolism	1	normal	352.55155	354.02645	831.01159

	variable	group	best_distribution	BIC_normal	BIC_lognormal	BIC_exponential
27	lipid_pero1	0	lognormal	2381.57583	2313.05191	3415.37440
28	lipid_pero1	1	lognormal	405.74102	402.53199	574.61506
29	lipid_pero2	0	lognormal	1703.22689	1645.88182	2622.25934
30	lipid_pero2	1	lognormal	308.28699	297.42006	443.59736
31	lipid_pero3	0	lognormal	871.73419	624.68815	1224.26787
32	lipid_pero3	1	lognormal	186.55714	161.29420	235.24911
33	lipid_pero4	0	lognormal	2379.81143	1928.59481	2689.54371
34	lipid_pero4	1	lognormal	352.00422	332.66036	453.80907
35	lipid_pero5	0	lognormal	5347.08685	5340.83564	6866.12298
36	lipid_pero5	1	normal	843.57680	889.85399	1113.90116
37	antioxidant1	0	normal	-473.29408	-433.74180	2934.53677
38	antioxidant1	1	normal	-115.40089	-113.12486	479.85869
39	antioxidant2	0	lognormal	1306.22766	1263.79179	3664.53808
40	antioxidant2	1	lognormal	206.39813	195.49228	591.67565

	variable	group	best_distribution	BIC_normal	BIC_lognormal	BIC_exponential
41	antioxidant3	0	normal	847.42297	855.42386	3325.63753
42	antioxidant3	1	lognormal	133.17594	132.70663	538.41113
43	antioxidant4	0	lognormal	5076.48748	4846.42823	6010.32102
44	antioxidant4	1	lognormal	826.95986	813.41265	1012.37846
45	antioxidant5	0	lognormal	375.33547	72.30235	1220.62608
46	antioxidant5	1	lognormal	64.29956	27.01913	238.77807

Table with descriptive statistics and specify the parameters according to the selected distribution:

R code:

```
final_table <- data.frame()
for (i in 1:nrow(results)) {
  var <- results$variable[i]
  grp <- results$group[i]
  dist <- results$best_distribution[i]
  subset_data <- data %>% filter(outcome == grp)
  x <- subset_data[[var]]
  mean_val <- mean(x, na.rm = TRUE)
  sd_val <- sd(x, na.rm = TRUE)
  median_val <- median(x, na.rm = TRUE)

  params <- get_params(x, dist)
  final_table <- rbind(final_table, data.frame(
```

```

    Variable = var,

    Group = grp,

    Distribution = dist,

    Mean = mean_val,

    SD = sd_val,

    Median = median_val,

    Param1 = as.numeric(params["param1"]),
    Param2 = as.numeric(params["param2"]),

    row.names = NULL
  ))
}

final_table <- final_table %>%
  mutate(across(where(is.numeric), round, 3))
print(final_table)

```

Output:

Variable	Group	Distribution	Mean	SD	Median	Param1	Param2
hormone1	0	lognormal	1.824	1.580	1.500	0.405	0.609
hormone1	1	lognormal	1.658	0.862	1.500	0.383	0.504
hormone3	0	lognormal	8.991	11.582	5.600	1.763	0.898
hormone3	1	lognormal	9.911	11.790	6.050	1.872	0.901
hormone4	0	lognormal	7.508	10.922	5.500	1.702	0.686
hormone4	1	lognormal	9.401	13.436	5.500	1.842	0.759

Variable	Group	Distribution	Mean	SD	Median	Param1	Param2
hormone5	0	lognormal	335.608	332.255	274.898	5.581	0.724
hormone5	1	lognormal	276.449	235.445	250.000	5.420	0.652
hormone6	0	lognormal	78.890	53.989	64.500	4.153	0.681
hormone6	1	lognormal	75.754	48.684	68.700	4.120	0.682
hormone7	0	lognormal	5.625	3.899	5.100	1.497	0.741
hormone7	1	lognormal	5.650	3.970	5.100	1.477	0.799
hormone8	0	lognormal	180.020	87.230	166.000	5.077	0.496
hormone8	1	lognormal	153.762	70.964	142.500	4.924	0.489
hormone10_generated	0	lognormal	0.762	8.298	0.427	-0.870	0.640
hormone10_generated	1	lognormal	0.508	0.366	0.456	-0.872	0.633
lipids1	0	lognormal	0.957	0.387	0.910	-0.125	0.412
lipids1	1	lognormal	0.985	0.370	0.980	-0.091	0.403
lipids2	0	lognormal	4.461	0.911	4.330	1.475	0.199
lipids2	1	lognormal	4.764	0.962	4.630	1.542	0.194

Variable	Group	Distribution	Mean	SD	Median	Param1	Param2
lipids3	0	lognormal	1.104	0.277	1.060	0.068	0.251
lipids3	1	lognormal	1.115	0.266	1.090	0.081	0.236
lipids4	0	lognormal	2.928	0.875	2.810	1.031	0.298
lipids4	1	lognormal	3.198	0.945	3.125	1.120	0.291
carb_metabolism	0	lognormal	4.895	0.807	4.800	1.576	0.157
carb_metabolism	1	normal	4.860	0.708	4.800	4.860	0.705
lipid_pero1	0	lognormal	2.068	0.803	1.970	0.649	0.405
lipid_pero1	1	lognormal	2.181	0.836	1.995	0.702	0.409
lipid_pero2	0	lognormal	1.384	0.570	1.300	0.237	0.436
lipid_pero2	1	lognormal	1.448	0.616	1.255	0.276	0.451
lipid_pero3	0	lognormal	0.682	0.374	0.590	-0.533	0.562
lipid_pero3	1	lognormal	0.755	0.421	0.640	-0.448	0.607
lipid_pero4	0	lognormal	1.432	0.803	1.320	0.236	0.504
lipid_pero4	1	lognormal	1.495	0.706	1.395	0.287	0.498

Variable	Group	Distribution	Mean	SD	Median	Param1	Param2
lipid_pero5	0	lognormal	11.879	3.609	11.660	2.426	0.318
lipid_pero5	1	normal	11.764	3.283	11.960	11.764	3.272
antioxidant1	0	normal	1.621	0.189	1.630	1.621	0.189
antioxidant1	1	normal	1.622	0.164	1.620	1.622	0.163
antioxidant2	0	lognormal	2.346	0.466	2.310	0.833	0.198
antioxidant2	1	lognormal	2.300	0.448	2.275	0.815	0.191
antioxidant3	0	normal	1.976	0.369	1.960	1.976	0.369
antioxidant3	1	lognormal	1.948	0.357	1.895	0.650	0.185
antioxidant4	0	lognormal	7.732	3.146	7.160	1.970	0.392
antioxidant4	1	lognormal	8.566	3.117	8.320	2.081	0.372
antioxidant5	0	lognormal	0.680	0.291	0.600	-0.467	0.397
antioxidant5	1	lognormal	0.764	0.287	0.710	-0.334	0.356